

Group Project: Graph Data Structure

1. Objective

This group project aims to help students deeply understand graph concepts, including nodes, edges, graph representation, traversal, shortest path, connectivity, and real-world modeling.

Each group must choose one (1) mini-project topic related to graph data structures and complete the required deliverables.

2. Group Formation

- Each group must have 3–5 members.
- Every member should contribute to research, slide preparation, and presentation.

3. Project Deliverables (Minimum Requirements)

A. Concept Presentation (Mandatory)

Each group must prepare a **PowerPoint slide deck (10–12 slides, but not limited to)** that includes:

1. Project Title & Group Members
2. Problem Background
3. Why Graph?
 - Show why the problem can be modeled as a graph.
4. Graph Design
 - Describe:
 - What your nodes represent
 - What your edges represent
 - Whether your graph is directed/undirected
 - Weighted/unweighted
 - Adjacency list or adjacency matrix
5. Algorithms Used
 - Example: BFS, DFS, Dijkstra, MST, cycle detection, etc.
6. Workflow Diagram or Pseudocode
7. Example Input & Expected Output

8. Limitations & Assumptions
9. Conclusion & Lessons Learned

Each group will give a **10-15 minutes presentation** in class.

4. Code Implementation (Bonus Score)

This part is optional but will give bonus points (**+10% to +20%**)

Bonus Requirements

If your group chooses to do a prototype:

1. Build a small **demo program** in any programming language.
2. The demo should:
 - Construct the graph
 - Show sample input
 - Run at least one algorithm (BFS/DFS/Dijkstra/MST etc.)
 - Print or visualize output

Note: Groups that do a code demo will present it after the slide presentation.

5. Submission Requirements

Each group must submit:

- PowerPoint slides (PDF)
- (Optional) Code folder or GitHub link
- Group member contribution statement (2–3 sentences each)

7. List of Suggested Mini-Projects

1. Campus Navigation System
2. Social Network Friend Recommendation
3. Airline Route Planner
4. Electricity Grid Simulation
5. Maze Solver

6. Internet Router Network
7. Movie Recommendation Graph
8. Disease Spread Simulation
9. Delivery Route Optimization
10. Timetable Conflict Checker (Graph Coloring)
11. Wikipedia Link Explorer
12. Kingdom Map Pathfinding